

Credit Quantity, Credit Prices, and Future Output

A model of heterogeneous financing access

Final presentation — work in progress

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`github.com/amchavezu/ai-project`

Question. Why can credit quantity be more informative for one sector, while the price of credit is more informative for another?

Answer. The relevant indicator depends on which financing margin determines the marginal unit of capital:

- ▶ at a binding cheap-bank cap, quantity determines feasible investment;
- ▶ with outside finance at the margin, its common price determines desired investment.

The thesis evidence is preliminary: the model explains a qualitative short-versus-longer-horizon pattern, not exact forecast peaks.

At date t , a firm with internal funds n chooses capital k , bank credit b , and outside finance m . Capital produces at $t + 1$.

$$\max_{k,b,m} \beta F(k) - R_B b - R_M m \quad \text{s.t.} \quad k = n + b + m, \quad 0 \leq b \leq \bar{b}, \quad m \geq 0.$$

$$F' > 0, \quad F'' < 0, \quad 0 < R_B \leq R_M < \beta F'(0).$$

Bank finance is cheaper but capped; the outside-finance wedge captures access, issuance, and information costs.

Let $\bar{k} = n + \bar{b}$, and define the two unconstrained targets by

$$\beta F'(k_B) = R_B, \quad \beta F'(k_M) = R_M, \quad k_M \leq k_B.$$

Then

$$k^* = \begin{cases} n, & k_B \leq n, \\ k_B, & n < k_B < \bar{k}, \\ \bar{k}, & k_M \leq \bar{k} \leq k_B, \\ k_M, & \bar{k} < k_M. \end{cases}$$

The empirical mechanism compares the third line (quantity regime) with the fourth line (price regime).

4 · Main result: two transmission margins

	Binding bank cap	Outside finance at margin
Strict regime	$k_M < \bar{k} < k_B$	$\bar{k} < k_M$
Capital	$k^* = n + \bar{b}$	$k^* = k_M(z, \tau)$
Bank-line effect	$\partial k^* / \partial \bar{b} = 1$	$\partial k^* / \partial \bar{b} = 0$
Spread effect	$\partial k^* / \partial z = 0$ locally	$\partial k^* / \partial z = 1 / [\beta F''(k_M)] < 0$
Economic signal	feasible investment pipeline	marginal cost of total finance

Zero effects are local: sufficiently large shocks can cross regime thresholds.

Let λ_s be the share of sector s in the quantity regime:

$$Y_{s,t+1} = \lambda_s F(n + \bar{b}_s) + (1 - \lambda_s) F(k_M(z_s, \tau_s)).$$

If $0 \leq \lambda_T < \lambda_N \leq 1$ and conditional primitives are common,

$$\frac{\partial Y_N}{\partial \bar{b}} > \frac{\partial Y_T}{\partial \bar{b}} \geq 0, \quad \left| \frac{\partial Y_N}{\partial z} \right| < \left| \frac{\partial Y_T}{\partial z} \right|.$$

This is a conditional composition result. Whether $\lambda_N > \lambda_T$ in Peru is an empirical assumption to test, not a theorem.

1. **Bank dependence:** quantity predictive content should rise with exhausted credit lines or dependence on relationship banks.
2. **Alternative funding:** spread predictive content should rise with access to bonds, foreign credit, or other external finance.
3. **Substitution:** a larger bank line should raise total investment at the cap, but mainly replace outside finance for market-access firms.

Project gestation and adjustment costs can make the price channel appear over a longer horizon. The present static proposition does not identify those dynamic parameters.

For $\beta = .95$, $a = 2$, $c = .5$, $n = .4$, $\bar{b} = .5$, $R_B = .5$, $R_M = 1.4$:

$$\bar{k} = .900, \quad k_B = 2.947, \quad k_M = 1.053.$$

Experiment	Quantity regime	Price regime
$\bar{b}$ rises by .10	capital $\uparrow$ to 1.000	capital unchanged
z rises by .02	capital unchanged	capital $\downarrow$ to 1.011

`code/verify.py` independently checks the FOCs, regime inequalities, comparative statics, and all reported values; it fails if a check is false.

- ▶ The hand appendix derives the policy and both propositions step by step.
- ▶ The current Lean 4 prototype proves the quadratic core and sectoral inequalities with no sorry.
- ▶ The required AppliedModelingLib-generated folder, pinned paper commit, and `check -fast` output remain pending for the final deadline.
- ▶ The paper appendix maps every numbered result to its Lean declaration.
- ▶ `prompts.md` preserves the raw AI exchanges and the paper records accepted, corrected, and rejected claims.

Known limits

- ▶ credit ceilings and sector composition are reduced-form primitives;
- ▶ the measured loan spread may not equal the marginal outside-finance cost;
- ▶ the forecasting relation is not a causal identification result.

Feedback sought

1. Is the two-tranche model the right minimum mechanism for the thesis?
2. Should the next extension endogenize the ceiling or model dynamics first?
3. Which Peruvian proxy best measures alternative-finance access by sector?